

THE PROFESSOR'S ACADEMY

XI PHYSICS MCQs (CHAPTERWISE)

CHAPTER # 2

- i. $K.(i \times j) =$
 * zero * **one** * $-j$ * $-k$
- ii. If two vectors of same magnitude, make an angle of 180° with each other than their resultant is:
 * $2F$ * $0.5 F$ * **Zero** * F
- iii. If a vector is dividend by its magnitude than the vector is:
 * **Unit vector** * Position vector * Null vector * Free vector
- iv. When $|A + B| = |A - B|$ then angle b/w them is:
 * 0° * 45° * **90°** * 180°
- v. The area of parallelogram formed by two vectors is:
 * $\frac{1}{2} A \cdot B$ * **$|A \times B|$** * $\frac{1}{2} |A \times B|$ * $A \cdot B$
- vi. The dot product of force and velocity is:
 * work * **power** * energy * momentum
- vii. Two perpendicular vectors having magnitudes 4 units and 3 units are added. The resultant has the magnitude:
 * 7 units * 12 units * **5 units** * 25 units
- viii. If $A \cdot B = 0$ and $A \times B = 0$ and $A \neq B = 0$ then $\vec{B}$ is:
 * equal to A * **zero** * perpendicular to A * parallel to A
- ix. $\sin \theta = \theta$ If θ is specifically less than:
 * 15° * 10° * **5°** * 1 Radian
- x. A vector which has a fixed reference point is called:
 * Unit vector * **Position vector** * Null vector * Free vector
- xi. The angle b/w i, j and k is:
 * 60° * **90°** * 120° * 180°
- xii. If two vectors of same magnitude combine together, they give resultant equal to double of any one, if angle b/w them is:
 * 0° * 60° * **90°** * 180°
- xiii. If $A \times B = 0$ whole $A \neq 0$ and $B \neq 0$ then:
 * **A and B are parallel** * A and B are perpendicular * A and B are opposite * None
- xiv. Two Non-zero vectors A and B are perpendicular if:
 * **$A \cdot B = 0$** * $A \cdot B = 1$ * $A \cdot B \neq 0$ * None
- xv. If $A \cdot B = 0$ when $A \neq 0$ and $R \neq 0$ then the vectors are:
 * parallel * opposite * **perpendicular** * both A and B
- xvi. If rectangular component of vectors 5m/sec in 3m/sec. What is the magnitude of 2nd component?
 • 2m/sec * 3m/sec * **4m/sec** * 5m/sec
- xvii. Law of cosine is normally used to determine the:
 * **magnitude of resultant** * direction of resultant * both magnitude and direction * None
- xviii. Law of sine is normally used to determine the:
 * magnitude of resultant * **direction of resultant** * both magnitude and direction * None
- xix. If the resultant of two forces of magnitude of 3N and 4N is 5N then the angle b/w the two forces is:
 * 0° * 45° * **90°** * 180°
- xx. The direction of vector is arbitrary:
 * **Null vector** * negative vector * positive vector * free vector
- xxi. Parallel vectors always have same:
 * **unit vector** * position vector * free vector * negative vector
- xxii. If rectangular components of a vectors are equal is magnitude, then the angle of vector from x-axis will be:
 * 30° * **45°** * 60° * 75°

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- xxiii. Additional of vectors obey _____ property.
 * Commutative * Associative * Distributive * **both a and b**
- xxiv. Two vectors of magnitude 10 units each making angles 30° and 90° from x-axis. The vertical component of resultant is:
 * 5 * 10 * **15** * 20
- xxv. Magnitude of sum and difference if two vectors are equal if the angle b/w the vectors is
 * 30° * 40° * 60° * **90°**
- xxvi. Magnitude of sum of two vectors is maximum only when the vectors are:
 * equal * **parallel** * antiparallel * perpendicular
- xxvii. Two forces of equal magnitude "F", the magnitude of their resultant is zero if vectors are _____ to each other.
 * perpendicular * parallel * **antiparallel** * None of them
- xxviii. Two forces of equal magnitude F, then the magnitude of their resultant is also F if angle b/w them is:
 * area of triangle * area of rectangle * area of parallelogram * area of circle
- xxix. Projection of any vector A along the direction of any other vector B may be calculated as:
 * $A \cdot B / |B|$ * $A \cdot B$ * $A \times B / |B|$ * **Both A and B**
- xxx. If a vector A lies in x y-plane and it makes an angle θ with y-axis, its y-component is:
 * **$A_y = A \cos\theta$** * $A_y = A \sec\theta$ * $A_y = A \tan\theta$ * $A_y = A \sin\theta$
- xxxi. A vector in space has _____ components.
 * one * two * three * None
- xxxii. The angle between the vectors $A = 2i + 3j + 6k$ and $B = 3i + 2j - 2k$ is:
 * 0° * 30° * **90°** * 180°
- xxxiii. If a force acting along y-axis is 5N, its component along x-axis is:
 * 5N * 10N * **0** * - 2N
- xxxiv. If the resultant of two vectors each of magnitude F, the angle between them will be:
 * 0° * 60° * 90° * **120°**
- xxxv. If $A = 3i + 4j$ then magnitude will be:
 * 7 * 25 * **5** * 1
- xxxvi. The cross product of vector A with itself ($A \times A$) is equal to:
 * A * 1 * A^2 * **0**

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CHAPTER # 3

1. A person walks 3 km towards east in 2hr and then returns to its initial position in 1 hr. The average velocity of the person is _____.
* 0 * **2 km/hr.** * 3 km/hr. * none of them
2. 1 m/sec = _____ km/hr.
* 0.500 * 0.2778 * **3.6** * 30
3. Value of gravitational acceleration in F.P.S system is _____ ft/sec².
* 9.8 * 10 * 16 * **32**
4. A boy throws a ball in upward direction while traveling in a bus with constant speed. The ball will drop _____.
* Behind the boy * before the boy
* **in the hand of the boy** * none of these
5. If we increase the momentum of the object twice, then the kinetic energy will be _____.
* same * doubled * Quartered * **quadrupled**
6. maximum static friction is also known as _____ friction.
* sliding * rolling * **limiting** * all of them
7. Inclined plane makes an angle θ which is _____.
* less than 90° * greater than 0°
* **both (A) and (B)** * none of them
8. As a result of constant un-balanced force, a body moves with _____.
* Uniform velocity * **uniform acceleration**
* variable speed * uniform speed
9. If the speed-time graph of a body is a straight line parallel to time –axis, its acceleration will be:
* **zero** * maximum * minimum * uniform
10. A 1kg stone when falling from a height of 10m, strikes the ground with velocity of _____.
* 10 m/s * **14 m/s** * 98 m/s * 196 m/s
11. A body goes from 2m to 8m mark and back to 2 m mark in 3 sec. Its average speed is _____.
* 2 m/s * 6 m/s * **4 m/s** * zero
12. If the speed-time graph of a moving body is a curve, the body moves with _____.
* constant speed * constant acceleration
* **constant velocity** * **changing acceleration**
13. The rate of change of linear momentum is equal to _____.
* acceleration * torque * angular velocity * **force**
14. When a constant force is applied on a body, it moves with a _____.
* constant speed * constant velocity
* **constant acceleration** * none
15. How many meter(s) will a 20 kg ball, starting from rest, fall freely in one second?
* 19.9 m * 9.8 m * 4.0 m * **4.9 m**
16. A helicopter weighting 3920N is moving up with a constant speed of 4m/sec. The force on the helicopter is.
* 4720N * **3920N** * 3924N * 3916N
17. The property of fluids due to which they resist their flow is called.
* static friction * coefficient of friction
* **viscosity** * terminal velocity
18. If the average and instantaneous velocities of a body are the same, the body will move with.
* variable velocity * **uniform velocity**
* uniform acceleration * variable acceleration
19. Under the influence of constant unbalanced force, a body moves with _____.
* uniform velocity * **uniform acceleration**
* variable acceleration * none of them

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20. A stone is thrown upward with 98m/sec. Time attain by the body is _____ to reach maximum height.
* **10 sec** * 15 sec * 20 sec * 25 sec
21. The product of force and time is called:
* Velocity * Acceleration * Power * **Momentum**
22. The acceleration of body moving down a frictionless inclined plane at 30° will be:
* **4.9m/sec^2** * 9.8m/sec^2 * 980m/sec^2 * 0
23. The unit momentum is:
* **N/sec** * N. sec * J/sec * J. sec
24. When a body slides over a surface then kinetic friction (f_k) and static friction (f_s) are related as:
* **$f_k < f_s$** * $f_k > f_s$ * $f_k = 0$ * $f_s = 0$
25. Momentum is conserved:
* In elastic collision * In inelastic collision * In explosion * **All of above**
26. When a small sphere travel down a viscous fluid, it moves with:
* Increasing velocity * Decreasing velocity * uniform acceleration * **uniform velocity**
27. A helicopter of mass 3000kg is moving with a constant speed of 25m/sec then resultant force on it is:
* 3×10^4 N * 7.5×10^4 N * 3000 N * **0**
28. A body is allowed to fall freely from a certain height, its speed at the end of first second will be:
* 4.9 m/sec * **9.8 m/sec** * 19.6 m/sec * 0
29. Two bodies of mass 1 kg and 4 kg moves with the same linear momentum, what will be relation b/w their velocities:
* 12 * 2 * **4** * 16
30. Swimming is possible due to which law of motion:
* 1st Law * 2nd Law * **3rd Law** * Law of gravitation
31. Whenever a constant force is applied on a body then it will move with:
* Constant speed * constant velocity * **constant acceleration** * None
32. If a particle is moving with constant speed in a circle than its velocity:
* zero * uniform * **variable** * instantaneous
33. 1 Newton = _____ Dyne
* **10^5** * 10^4 * 10^3 * 10^2
34. If force is acting on a body is doubled then acceleration produced in:
* halved * **doubled** * one forth * same
35. If a body is moving with constant velocity, then:
* Its direction may be changing * Its acceleration is variable
* **Its acceleration is zero** * None of these
36. If a body is moving along a circle with constant speed, then:
* Its velocity is uniform * **Its velocity is changing**
* Its acceleration is increasing * Its acceleration is zero
37. A body whose momentum is constant, must have constant:
* **Velocity** * force * acceleration * none of these
38. The acceleration of a spherical ball on a smooth inclined plane is maximum when the angle of inclination to the horizontal:
* **90°** * 60° * 30° * 0°
39. When a force of one 4N act on a mass of 2kg for a time of 2s, what is the rate of change of momentum:
* 1 kg. m/s² * 2 kg m/s² * **4 k m/s²** * 8 kg m/s²
40. If a body is at rest, it may have:
* momentum * velocity * acceleration * **energy**

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CHAPTER # 4

1. The path of projectile:
* **Parabola** * Hyperbola * Circular * Elliptical
2. During the projectile motion the acceleration along the horizontal direction will:
* Decrease * Increase * **Be zero** * remain same
3. In projectile motion the acceleration along the vertical direction will:
* Decrease * Increase * **remain same** * Be zero
4. If the angle of elevation of projectile is 90° then its horizontal range will be:
* Minimum * Maximum * **Zero** * None of these
5. The range of projectile become half of its maximum value of the angle of projectile is:
* **15°** * 30° * 45° * 60°
6. The horizontal component of velocity during projectile motion always:
* zero * **remains same** * increases * decreases
7. If the initial speed of projectile is doubled its range will:
* Decrease by one half * **Increase four times** * Doubled
8. When an object moves with constant speed in a circle it has:
* **centripetal acceleration only** * tangential acceleration
* both centripetal and tangential acceleration * None
9. The range of projectile is maximum when the angle of projection is:
* 0° * 30° * **45°** * 60°
10. A body is moving along a circle with increasing speed., It has:
* tangential acceleration * centripetal acceleration
* **both** * no acceleration
11. When a body moves with constant speed in a circle then:
* **Its velocity is changing** * Its acceleration is zero
* Its velocity is uniform * Its acceleration is increasing
12. If a projectile is launched at 45° with velocity of 100m/sec, it will have double the range of its velocity is:
* 141.4 m/sec * **200 m/sec** * 400 m/sec * zero
13. At maximum height the vertical component of velocity is:
* **zero** * maximum * uniform * remains same
14. If the velocity of body moving on circular path is doubled then its centripetal acceleration becomes:
* $1/2$ * $1/4$ * double * **four times**
15. During circular motion the angle between linear and angular velocity is always:
* 0° * **90°** * 180° * 2π radian
16. The angular speed of seconds hand of watch is:
* **$\frac{\pi}{30}$ rad/sec** * $\frac{\pi}{60}$ rad/sec * π rad/sec * 1 rad/sec
17. If the angular velocity decreases, then the direction of angular acceleration will:
* **opposite to one another** * perpendicular to one another
* same as one another * None
18. If the angular velocity decreases, then the direction of angular acceleration will:
* **opposite to one another** * perpendicular to one another
* same as one another * none
19. The angle subtended at the centre by a body moving along a circle is called:
* Angular speed * Angular velocity
* **Angular displacement** * Angular acceleration
20. A point on the rim of wheel moves 0.2m when the wheel turns through an angle of 0.1 radian. What is the radius of wheel?
* 0.5m * **2m** * 0.2m * 20m

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21. A bomber drops the bomb when it is vertically above the target, it misses because of:
 * velocity * acceleration * vertical velocity * **horizontal velocity**
22. If the initial speed of projectile is doubled, its range will:
 * decrease by one half * **increases 4 times**
 * one forth * double
23. In projectile the acceleration along vertical direction will:
 * decrease * increase * **remains same** * zero
24. The expression for the time period of an object moving with constant speed "V" along a circle of radius "r" is given by:
 * $4\pi r/V$ * **$2\pi r/V$** * $\pi r^2/V$ * $4\pi r^2/V$
25. The direction of linear velocity of a particle moving along a circle is directed:
 * away from centre * towards centre
 * **tangent to the circle** * none
26. We get the same range of projectile at two angles given by:
 * **$15^\circ, 75^\circ$** * $30^\circ, 40^\circ$ * $40^\circ, 60^\circ$ * $25^\circ, 75^\circ$
27. During upward motion of the projectile the vertical component of velocity:
 * **decreases** * increases * remains same * none
28. The force responsible for the circular motion of electron around the nucleus is:
 * frictional force * Gravitational force * **Coulomb's force** * Cohesive force
29. 1 Radian = _____ :
 * 37.5° * **57.3°** * 53.7° * 90°
30. When an object moves along a circular path its velocity:
 * **remains constant** * changes continuously * becomes zero * none of these
31. When a particle moves in a circle, the angle between linear and angular velocity is:
 * 0° * **90°** * 180° * None
32. The max. height of projectile depends upon:
 * Angle of projection * Velocity of projection
 * **Both angle and velocity** * None
33. When a body moves along circumference of circle with uniform speed, changes take place in its:
 * **Linear velocity** * Tangential acceleration * Both * None
34. When a body moves along parabolic path, which component of velocity does not change:
 * **Horizontal** * Vertical * Both horizontal and velocity * None
35. The motion on a curved path when one component of velocity is constant and other is variable is called:
 * Circular motion * **Projectile motion** * Vibratory motion * All of these
36. Centripetal force is also called:
 * **centrifugal force** * centre-seeking force * tangential force * none of these
37. A cyclist cycling around a circular track skid because:
 * **The centripetal force upon him is less than the limiting friction**
 * The centripetal force upon him is greater than the limiting friction
 * The centripetal force upon him is equal to the limiting friction
 * None of these
38. When the angular velocity of disc increases, then angular velocity and angular acceleration are:
 * **parallel** * non-parallel * perpendicular * opposite
39. The angle subtended at its centre by an arc whose length is double to that of its radius is:
 * 57.3° * **114.6°** * 90° * 180°
40. The angular speed of minute hand clock is:
 * **$\pi/30$** * $\pi/60$ * $\pi/1800$ * $\pi/3600$
41. A body along the circumference of circle complete two revolutions. If the radius of circle is R, the ratio of its displacement to the covered path will be:
 * zero * πr * $2\pi r$ * **$4\pi r$**

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42. In projectile motion the acceleration along vertical direction will:
* decrease * increase * **remains the same** * be zero
43. During projectile motion the acceleration along the horizontal direction will:
* remains same * **be zero** * decrease * increase
44. In projectile motion the small angle of elevation produces:
* Flat trajectory * **high trajectory** * low trajectory
45. For projection of low trajectory, their time of flight will be:
* **short** * long * none of these
46. If an object is velocity in the counterclockwise direction, then the direction of angular velocity is:
* into the paper * **out of the plane** * both of these * None
47. If a particle moving with constant speed along the circumference of a circle, then the acceleration is:
* Linear acceleration * **Centripetal** * Tangential * All of these
48. When an object moves round the circular track, the centripetal force is provided by:
* force of gravity * **frictional force** * centrifugal force * All of these
49. At _____ angle the horizontal range becomes twice as max. height for a projectile:
* $\tan^{-1}(1)$ * **$\tan^{-1}(2)$** * $\tan^{-1}(3)$ * $\tan^{-1}(4)$
50. Maximum horizontal range is always _____ timing the maximum height of projectile.
* two * three * **four** * None
51. In uniform circular motion _____ remains constant:
* speed * angular speed * acceleration * **All of them**
52. The angular speed of minute needle of clock is _____ rad/sec.
* $\pi/30$ * $\pi/60$ * **$\pi/1800$** * $\pi/21600$
53. No. of revolutions and angular displacement are related as:
* **$\theta = 2n\pi$** * $n = 2\pi\theta$ * $\theta = 2\pi/n$ * None

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CHAPTER # 5

1. The unit of angular momentum is:
* N. sec * **J. sec** * N.m * None of these
2. Two forces which are equal in magnitude, but opposite in direction constitute a _____:
* Circle * **Couple** * Momentum * Power
3. The moment of force depends on:
* Force * Position * Choice of origin * **All of them**
4. Quantity of angular motion is:
* Angular displacement * Angular Speed
* Angular acceleration * **Angular momentum**
5. Earth has _____ angular momentum:
* Spin * Orbital
* **Both spin and orbital** * Neither spin nor orbital
6. In line of action of force passes through axis of rotation, moment arm is equal to:
* r * **zero** * $\frac{1}{2} r$ * r^2
7. The perpendicular distance between line of action of force and axis of rotation is called:
* Displacement * Distance
* **Moment arm** * Angular displacement
8. A body will be in rotational equilibrium if:
* $\sum F = 0$ * $\sum L = 0$ * **$\sum T = 0$** * $\sum r = 0$
9. A body satisfying the 2nd condition of equilibrium will be in:
* translational equilibrium * **rotational equilibrium**
* neutral equilibrium * complete equilibrium
10. A paratrooper falling towards the earth is in state of equilibrium because:
* He experiences no gravitational force * The air is not blowing
* **Force of gravity is balanced by Gravitational force** * He is falling vertically downward
11. The vector quantity torque:
* **depends on the choice of origin** * doesn't depend on the choice of origin
* rotates a body only in clockwise
12. The direction of torque is:
* the same as the direction of the corresponding applied force
* opposite to the direction of the applied force
* **perpendicular to the direction of applied force**
13. The moment of momentum is called:
* Couple * torque * **Angular momentum**
14. If the distance of a particle from the axis of rotation is doubled, the moment of inertia:
* becomes half * increases two times
* **increases four times** * increases eight times
15. In rotational motion, the analog of force is:
* rotational inertia * moment of inertia * **torque**
16. The term torque is synonymous with:
* **moment of force** * moment of inertia * angular momentum
17. If the direction of the applied force is reversed, then:
* its torque remains unchanged
* the magnitude of its torque changes and direction of the torque remains the same
* **the magnitude of its torque remains the same but the direction of the torque reverses**
18. The arm of couple is:
* the smallest distance between two equal and antiparallel forces
* the greatest distance between two equal and antiparallel forces
* **the lines of action of two equal and opposite forces**

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19. The angular momentum (L) can be expressed in terms of moment of inertia (I) and angular velocity (ω):
* **$L = I\omega$**
* $L = I\omega^2$
* $L = I/\omega$
20. It is easier to turn a steering wheel with both hands than with a single hand because:
* **a couple acts on the wheel** * two equal and opposite forces act on the wheel
* the wheel is more strongly gripped
21. The physical quantity which produces angular acceleration is called:
* **torque** * work * power * force
22. Two forces which are equal in magnitude but opposite in direction and are not acting along the same line constitute a:
* **couple** * circle * power * force
23. The unit of angular momentum is:
* N.s * **J.s** * N.m * none of these
24. When the net torque acting on the system is zero which of the following will be constant:
* Force * **Angular momentum** * Linear momentum * None of these
25. The rate of change of angular momentum is also called:
* Force * Acceleration * Initial * **Torque**
26. The angular momentum of a particle is conserved if the:
* **Net torque on the particle is zero** * Net force on the particle is zero
* The acceleration of the particle is zero * Net displacement of the particle is zero
27. If the axis of rotation of a rotating body passes through the body itself, then its motion is called:
* **Linear motion** * Orbital motion * Spin motion * S.H motion
28. The magnitude of torque of the couple depends on:
* The distance of (F) from the origin * The distance of (-F) from the origin
* **The distance between (F) and (-F)** * None of these
29. Torque is maximum when force:
* is parallel to moment arm * is anti-parallel to moment arm
* makes an angle of 60° with moment arm * **is perpendicular to moment arm**
30. The ratio of S.I unit of Angular momentum to linear momentum is:
* J.s * $N.J^{-1}$ * J.N * **$J.N^{-1}$**
31. A force of 8N is applied to the spanner perpendicularly at a distance of 0.15m from the center of the nut, the moment of the force acting on the nut is:
* **1.2 N.m** * 1.5 N.m * 2.1 N.m * 3 N.m
32. The point which describes the motion of the whole system or body is known as the:
* inertia * center of gravity * **center of mass** * moment of inertia
33. A particle moving with uniform speed in circle has _____ acceleration:
* zero * maximum * **uniform** * none of these

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CHAPTER # 6

1. Inertial mass and Gravitational mass are _____.
(a) Opposite (b) Identical
(c) Weights (d) **Proportional**
2. Average density of earth is
(a) $55 \times 10^3 \text{ Kg/m}^3$ (b) **$5.5 \times 10^3 \text{ Kg/m}^3$**
(c) 2.25×10^3 (d) 2.55×10^{13}
3. If earth is supposed to be a perfect square, then the value of g will be
(a) Grater at the equator (b) Constant at all the places
(c) **Least at the equator** (d) None of these
4. The ratio of the dimension of G to that of g is _____.
(a) L^3M^{-2} (b) LM^{-2}
(c) L^2M^{-2} (d) **L^2M^{-1}**
5. If a planet whose mass and radius both twice that of earth than acceleration due to gravity will be
(a) **4.9 m/sec^2** (b) 9.8 m/sec^2
(c) 19.6 m/sec^2 (d) 2.45 m/sec^2
6. If the radius of earth has to shrink by 1% while mass remains same, then the acceleration due to gravity on the earth surface will
(a) Decrease (b) **Increase**
(c) Remains same (d) None of these
7. What happens to the person's weight when he goes down to deep mine _____?
(a) It will remain same (b) It will increase
(c) **It will decrease** (d) 1st increase then decrease
8. The weight of an object at the pole is greater than at equator, this is because;
(a) **The gravitational pull is more at the poles** (b) The shape of earth
(c) The attraction of moon is maximum at earth (d) None
9. On the surface of moon the weight of a person
(a) Increases (b) **Decreases**
(c) Remains same (d) None
10. A person whose weight is 120 pounds on earth, the weight on moon will be
(a) 20 pounds (b) 30 pounds
(c) 40 pounds (d) **120 pounds**
11. Acceleration of moon is about _____ m/sec^2
(a) **2.72×10^{-3}** (b) 2.72×10^3
(c) 9.8 (d) 2.7×10^4
12. The value of radius of moon is about;
(a) **$3.84 \times 10^8 \text{ m}$** (b) $3.84 \times 10^5 \text{ m}$
(c) $3.84 \times 10^3 \text{ m}$ (d) $3.84 \times 10^{10} \text{ m}$
13. The earth completes its circular orbit in 3.15×10^7 sec with an orbital velocity of $2.9 \times 10^4 \text{ m/sec}$, then its orbital radius in;
(a) **$1.45 \times 10^{11} \text{ m}$** (b) $1.45 \times 10^8 \text{ m}$
(c) $1.45 \times 10^6 \text{ m}$ (d) $1.45 \times 10^4 \text{ m}$
14. The mass of planet and its diameter are three times those of earth then the acceleration due to gravity on the surface of planet will be
(a) **One third** (b) Half
(c) One ninth (d) None
15. If the distance b/w the center of body to center of earth increases, then the value of g _____.
(a) Increases (b) **Decreases**
(c) Remains the same (d) None

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16. The apparent weight of a body in a satellite orbiting around the earth is always;
(a) **Zero** (b) Mg
(c) Increasing (d) Decreasing
17. Above the surface of earth if we go to the distance equal to double of its radius of earth then the value of "g" will become;
(a) **One ninth** (b) One third
(c) One fourth (d) One half
18. The ocean tides are caused by gravitational force exerted on earth by;
(a) Moon only (b) Sun only
(c) **Both moon and sun** (d) None
19. If we go up from the surface of earth to a distance equal to radius of earth, the value of "g" will be
(a) $\frac{1}{2}g$ (b) **$\frac{1}{4}g$**
(c) $2g$ (d) $4g$
20. The value of gravitational constant "g" was determined experimentally by;
(a) **Cavendish** (b) **Newton**
(c) Joule (d) Faraday
21. As compared to the density of surface layer, the density of interior of earth is
(a) **Lesser** (b) Greater
(c) Equal (d) Zero
22. The value of "g" is greater at;
(a) **Poles** (b) Equator
(c) Both A and B (d) None
23. The minimum velocity at which body must be projected to go out from earth" gravitational pull is called;
(a) Angular velocity (b) **Escape velocity**
(c) Terminal velocity (d) All of these
24. At what depth below the earth surface the value of "g" reduces to one half of its value on earth surface:
(a) **$\frac{R}{2}$** (b) $\frac{R}{4}$
(c) R (d) $2R$
25. The sun exerts gravitational force on the planets to keep them into their;
(a) Radius (b) Axes
(c) Motion (d) **Orbit**
26. The value of "g" at center of earth is
(a) 9.8 m/sec^2 (b) Infinity
(c) **Zero** (d) Double
27. As the depth from the surface of earth is increases the value of "g".
(a) **Decreases** (b) Increases
(c) Remains same (d) None of above
28. The weight of body of mass 10kg at center of earth will be
(a) 980 N (b) **Zero**
(c) 1960 N (d) 490 N
29. If mass of stone is 80g then its weight is
(a) 256 dynes (b) 8N
(c) **78400 dynes** (d) 80 dynes
30. Which one of the following is long range force in nature
(a) **Gravitational force** (b) Weak nuclear force
(c) Strong nuclear force (d) Electrostatic force

THE PROFESSOR'S ACADEMY

XI PHYSICS MCQs (CHAPTERWISE)

CHAPTER # 7

1. A person of 70 kg lifts a mass 30kg to the roof 10m high. What power spent by him in 50 seconds?
(a) 19.6 W (b) **196 watts**
(c) 600 watts (d) 800 watts
2. Sum of K.E and P.E is referred as;
(a) Combine energy (b) Total energy
(c) **Mechanical energy** (d) Fundamental energy
3. Formula of K.E is
(a) $\frac{1}{2}mv^2$ (b) $\frac{p^2}{2m}$
(c) $\frac{2p^2}{2m}$ (d) **Both A and B**
4. A body is dragged on a horizontal surface, its P.E;
(a) **Remains constant** (b) Increases
(c) Decreases (d) None of these
5. A force is applied in lifting 100 kg mass to a height of 10m, the work done is
(a) **9800 J** (b) 500 J
(c) 1000 J (d) 19.6 J
6. The scalar product of force and velocity is
(a) **Power** (b) Work
(c) Momentum (d) Energy
7. If weight of body is 600N and it is taken up to 10m height in 10 sec, then power is
(a) 6000 watts (b) **600 watts**
(c) 60 watts (d) 6 watts
8. In British Engineering system the unit of power is
(a) Watt (b) Kilowatt
(c) Foot watt (d) **Horsepower**
9. If the velocity of body is doubled then its K.E becomes
(a) One – fourth (b) Double
(c) One half (d) **4 times**
10. 1 Joule = _____
(a) $\frac{1N}{1m}$ (b) $\frac{1m}{1N}$
(c) $\frac{1N}{1m^2}$ (d) **1N.1m**
11. 1 Joule = _____ Ergs.
(a) 10^5 (b) 10^3
(c) **10^7** (d) 10^8
12. K.E of a body depends upon its _____.
(a) Position (b) Velocity
(c) Mass (d) **Mass and velocity**
13. The energy of water in tank or dam is
(a) Kinetic (b) **Potential**
(c) Gravitational (d) Electrical
14. 33.5 hp = _____ watt
(a) 5000 watts (b) 2500
(c) **25000** (d) 250

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XI PHYSICS MCQs (CHAPTERWISE)

15. The work done by centripetal acceleration is always;
(a) Positive (b) **Zero**
(c) Negative (d) All these
16. 1HP = _____ watt;
(a) 546 (b) **746**
(c) 760 (d) 646
17. Which one is not a unit of power
(a) Kilowatt (b) **Kilowatt hour**
(c) Horsepower (d) Ft-lb/sec
18. In B.E system 1HP = _____.
(a) **550 ft.lb/sec** (b) 746 ft.lb/sec
(c) 1055 ft.lb/sec (d) 50 ft.lb/sec
19. 1Kwh = _____ J
(a) 360 J (b) 3600 J
(c) 360000 J (d) **3600000 J**
20. Watt hour is the unit of;
(a) Force (b) Acceleration
(c) **Energy** (d) Velocity
21. A weightlifter consumes 500J of Energy to lift a load in 2 seconds, the power used by him is
(a) 125 watts (b) **250 watts**
(c) 500 watts (d) 1000 watt
22. Both Kw/h and electron volt (EU) are the unit of
(a) Power (b) **Energy**
(c) Angular momentum (d) Momentum
23. A man weighting 600N climbs 5m vertically upward in 8 sec, his rate of working is
(a) **175 watts** (b) 275 watts
(c) 375 watts (d) 975 watts
24. The dimensions of K.E are
(a) **$ML^2 T^{-2}$** (b) $M L^{-1} T^{-2}$
(c) $M^{-1} L^2 T^{-2}$ (d) $M^2 LT^2$
25. Which quantity has similar dimensions?
(a) Work and power (b) Power and K.E
(c) **Work and K.E** (d) P.E and Power

THE PROFESSOR'S ACADEMY

XI PHYSICS MCQs (CHAPTERWISE)

CHAPTER # 8

1. A stretched string b/w two fixed points is vibrating in one segment. The frequency generated is called
 - (a) 1st overtone
 - (b) **Fundamental frequency**
 - (c) Normal harmonic
 - (d) None
2. The loudness of sound depends upon
 - (a) Wavelength
 - (b) Frequency
 - (c) **Amplitude**
 - (d) All these
3. The sound waves are
 - (a) **Longitudinal**
 - (b) Transverse
 - (c) Medium is not necessary
 - (d) None of these
4. Maximum beat frequency which human ear can detect is
 - (a) 3
 - (b) 5
 - (c) **7**
 - (d) 9
5. If a string is vibrating its fundamental mode, then to excite the next higher mode the tension must be
 - (a) Halved
 - (b) Doubled
 - (c) Tripled
 - (d) **Quadrupled**
6. If the mass of bob of a simple pendulum is replaced by circular motion, then its time;
 - (a) Increases
 - (b) Decreases
 - (c) **Remains the same**
 - (d) Becomes infinite
7. Speed of sound waves have the frequency of mass than;
 - (a) 20 Hz
 - (b) 2000 Hz
 - (c) 5000 Hz
 - (d) **20,000 Hz**
8. The intensity level of sound of intensity 10^{-12} watt/m² in bell is
 - (a) **Zero**
 - (b) 60
 - (c) 120
 - (d) None
9. Which of the following property of sound is affected by change in temperature?
 - (a) Amplitude
 - (b) **Wavelength**
 - (c) Frequency
 - (d) Intensity
10. If the mass of body is suspended from a spring is doubled then the period of vibration of body becomes
 - (a) Double
 - (b) Half
 - (c) **$\sqrt{2}$ times**
 - (d) Four times
11. The S.I unit of intensity of sound is;
 - (a) **$\frac{\text{watt}}{\text{m}^2}$**
 - (b) Decibel
 - (c) Weber
 - (d) Diopter
12. Power law determines
 - (a) Power
 - (b) Work
 - (c) Intensity
 - (d) **Loudness of sound**
13. A wave enters from one medium to another medium, no change occurs in its:
 - (a) **Frequency**
 - (b) Wavelength
 - (c) Amplitude
 - (d) Speed
14. The time period of simple pendulum depends upon;
 - (a) Mass
 - (b) Length
 - (c) Acceleration due to gravity
 - (d) **Both length and acceleration**
15. The product of frequency and time period is
 - (a) **1**
 - (b) 2
 - (c) 3
 - (d) 4
16. The time period of simple pendulum at the center of earth is
 - (a) Zero
 - (b) Unity
 - (c) **Infinity**
 - (d) Same as that on surface of earth

THE PROFESSOR'S ACADEMY

XI PHYSICS MCQs (CHAPTERWISE)

17. When pressure of the gas is doubled the velocities of sound in it are;
(a) Doubled (b) Halved
(c) Squared (d) **Unaltered**
18. When the temp of gas is increased to 4 times the velocity of sound V becomes;
(a) $V/4$ (b) $V/2$
(c) **$2V$** (d) $4V$
19. The intensity of one sound is 10 times the intensity of other sound, the two sounds differ by:
(a) 1 decibel (b) **1 bel**
(c) 10 decibels (d) 100 decibels
20. The speed of sound in air at 260 K is 320 m/sec the speed of sound in air at 310 K is
(a) 270 m/sec (b) 290 m/sec
(c) 320 m/sec (d) **350 m/sec**
21. The speed of sound is determined by its
(a) Amplitude (b) Intensity
(c) **Pitch** (d) Number of overtones
22. The length of second pendulum where $g = 980 \text{ cm/sec}^2$ is:
(a) **100 cm** (b) 900 cm
(c) 490 cm (d) 980 cm
23. If mass of simple pendulum is doubled its time period;
(a) Becomes double (b) Becomes half
(c) One forth (d) **Remain same**
24. The waves which require a material medium for their propagation are called;
(a) Electromagnetic wave (b) **Mechanical wave**
(c) Matter waves (d) Carrier waves
25. Which of the following waves are transverse waves
(a) Sound waves (b) Spring waves
(c) **Water waves** (d) All of them
26. The velocity of longitudinal waves in a metallic rod depends upon;
(a) Bulk modulus (b) **Young's modulus**
(c) Stress (d) Rigidity
27. Doppler effect can be applied to;
(a) Light waves (b) Sound waves
(c) **Both light and sound waves** (d) None of the above
28. Wave that does not require any medium to propagate is known as _____ wave;
(a) **Electromagnetic** (b) Mechanical
(c) Transverse (d) Longitudinal
29. Such waves in which particles of the medium execute S.H.M is known as;
(a) **Sinusoidal waves** (b) Mechanical wave
(c) Transverse waves (d) Longitudinal waves
30. Number of waves per cycle is called;
(a) Amplitude (b) Frequency
(c) Wavelength (d) **Wave number**
31. Which is not the transverse wave;
(a) Wave on string (b) Water waves
(c) Light waves (d) **Sound waves**
32. Longitudinal waves cannot be
(a) Reflected (b) Interfered
(c) **Polarized** (d) All of the above
33. According to Newton sound travels ;
(a) **Isothermally** (b) Isobarically
(c) Adiabatically (d) Isochronally

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XI PHYSICS MCQs (CHAPTERWISE)

34. According to Laplace sound travels;
(a) **Adiabatically** (b) isobarically
(c) Isothermally (d) All of above
35. Supersonic object like F-16 jet plane produces _____ waves which travelling in air;
(a) **Shock waves** (b) Electromagnetic waves
(c) Standing waves (d) Sinusoidal waves
36. A body is executing S.H.M at the mean position it's;
(a) Acceleration is maximum (b) **Acceleration is zero**
(c) Velocity is zero (d) Possess max. potential energy
37. A simple pendulum is in S.H.M with period T. If length is doubled, the new time period will be
(a) 2T (b) 0.5 T
(c) 2.5 T (d) **1.414 T**
38. If we increase the length of simple pendulum its time period will;
(a) **Increases** (b) Decreases
(c) Remains same (d) Becomes infinite
39. When the temperature of air rises, the speed of sound waves increases because;
(a) Frequency of wave increases (b) **Wavelength increases**
(c) Both frequency and wavelength increase (d) None
40. A simple pendulum is transported to moon its frequency of oscillation will
(a) **Decreases** (b) Increase
(c) Remains constant (d) Becomes zero
41. For a system to execute S.H.M, it must possess
(a) Elasticity (b) Inertia
(c) **Both** (d) None of these
42. The S.I unit of force constant is identical to that of
(a) Pressure (b) Energy
(c) **Surface tension** (d) Force
43. In S.H.M graph b/w force and displacement is
(a) Parabolic (b) Hyperbolic
(c) Exponential (d) **Linear**
44. A source of sound wave moves away with the velocity of sound from a stationary observer. The frequency of note is
(a) Unchanged (b) Doubled
(c) **Halved** (d) Squared
45. Sound travels faster in;
(a) Air (b) Water
(c) Vacuum (d) **Glass**
46. When sound waves travel from air to water, the quantity that remains unchanged is;
(a) Speed (b) Frequency
(c) Intensity (d) **Wavelength**
47. When a pressure of gas is doubled the velocity of sound in it is
(a) Doubled (b) Halved
(c) **Unaltered** (d) Squared
48. At what temperature with the velocity of sound be double of its value at 0°C.
(a) 81.9°C (b) **819°C**
(c) 8190°C (d) None of these
49. Damping is
(a) Reduction in frequency (b) Reduction in wavelength
(c) **Reduction in amplitude** (d) All of above

THE PROFESSOR'S ACADEMY

XI PHYSICS MCQs (CHAPTERWISE)

50. The amplitude of sound wave determines its
(a) **Loudness** (b) Pitch
(c) Quality (d) Interference
51. The loudness of sound depends upon _____.
* 1 overtone * Fundamental frequency * Normal harmonic * **none of these**
52. When the temperature of air raises the speed of sound increases because.
* Frequency of wave increases * Both the frequency and the wavelength increases
* Only wavelength increases * None of these
53. When a sound source moves towards a stationary listener, there is _____.
* An apparent increase in wavelength * **An apparent increase in frequency**
* A decrease in pitch * None of these
54. The sound waves are _____.
* **Longitudinal** * Transverse * both A and B * none of these
55. In aerodynamics the ratio of the velocity of the sound waves to the velocity the sound source is called _____.
* Beats * **Mach number** * Harmonics * none of these
56. When the speed of jet plane becomes greater than the speed of sound then loud sound produced is due to _____.
* **Shock waves** * Standing waves * none of these
57. Pitch of sound depends upon _____.
* Intensity * Loudness * **Frequency** * None of these
58. Which of the following represents compressional waves?
* **Sound waves** * Light waves * Radio waves * X-rays
59. The distance between the nodes and anti-nodes in a loop is _____.
* $\frac{\lambda}{4}$ * $\frac{\lambda}{2}$ * $\frac{\lambda}{3}$ * none of these
60. A decibel is a _____.
* Musical instrument * **Measure of intensity level** * Wavelength * Musical note
61. Speed of sound waves does not depend upon _____.
* Wind speed * Temperature * **Pressure** * Density of the medium
62. The acceleration of free fall on the Moon is $\frac{1}{6}$ th of that on the Earth. What would be the period on the Moon of a simple pendulum which has a period of 1 second on the Earth?
* $\frac{1}{6}$ the second * $\frac{1}{\sqrt{6}}$ the second * **$\sqrt{6}$ th second** * 6 second
63. When the temperature of air rises, the speed of sound wave increases because.
* Only frequency increases * **Only wavelength increases**
* Both wavelength and frequency increases * Only wavelength decreases.
64. The frequency of oscillation of a simple pendulum depends upon _____.
* Mass of the bob * Amplitude of vibration * **length of pendulum** * none of these
65. The relative intensity $\left(\frac{1}{I_i}\right)$ for the sound of a jet engine is 10^{13} . The intensity level in dB will be _____.
* 13 * 1300 * **130** * 1.3
66. The intensity level of sound of waves travels in a medium in opposite directions, they produce _____.
* Zero * One * Two * three
67. When two exactly similar waves travel in a medium in opposite directions, they produce _____.
* **Standing waves** * Beats * Resonance * Diffraction
68. The physical quantity which is related to the loudness of sound is _____.
* Frequency * **Intensity** * Quality * Temperature

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XI PHYSICS MCQs (CHAPTERWISE)

69. Which of the following properties of sound is affected by change in temperature?
 *Amplitude ***Wavelength** *Frequency * Intensity
70. If the mass of the bob of a simple pendulum is doubled, its time period will _____.
 *Be doubled * Be Half ***Remains constant** * none of these
71. The frequency of a second's pendulum is _____.
 * 1 Hz *2 Hz ***0.5 Hz** * none of these
72. Which of the following is a compressional wave?
 *Light wave *** Sound wave** * Radio wave * none of these
73. If the bob of vibrating pendulum is suddenly detached from the siring at its mean position, its;
 *A straight line *A circle ***A parabola** * A hyperbola
74. The velocity of sound in a gas increases with _____.
***Temperature** *Pressure * Loudness * Frequency
75. Which of the following is not the property of sound waves?
 *interference * diffraction ***Polarization** * Refraction
76. "Sone" is the unit of _____.
 *Intensity level * Intensity of sound *** Loudness of sound** * Quality of sound
77. Which of the following represents longitudinal waves?
 *Light waves ***Sound waves** *Radio waves * X-rays
78. One Sone at 1000 Hz is equal to _____.
 *60 dB *** 40 dB** *30dB * 100 dB
79. The Earthquake waves are the examples of _____.
 *Audible waves *** Infrasonic waves** *Shock waves * Ultrasonic waves
80. When the temperature of air rises, the speed of sound increases because _____.
 *The frequency of the wave increases
***The wavelength of the wave increases**
 *Both the wavelength & frequency increases
 * Neither the wavelength nor the frequency increases
81. If you go on increasing the stretching force on a guitar its frequency of vibration _____.
***Increases** * Decreases *neither decrease nor decreases * becomes zero
82. Which of the following does not exhibit simple harmonic motion?
 *A hanging spring supporting a weight
 *The balance wheel of a watch
***The wheel of an automobile**
 *The string of a violin
83. The velocity of sound in space is _____.
 *332 m/sec *334 m/sec *A 330 m/sec *** Zero**
84. This is compressional waves.
 *Light waves ***Sound waves** *Radio waves * X-Rays
85. The S.I unit of intensity level of sound is.
 *Watt *Dioptr *Sone *** Decibel**
86. The frequency of wave produced in a stretched string depends upon.
 *Length *Tension *Linear density *** All of these**
87. The distance between two consecutive nodes of a stationary wave will be.
 * λ *** $\lambda / 2$** * $\lambda / 4$ * $\lambda / 6$
88. Motion that repeats itself in equal interval of time is called _____.
 *Harmonic motion *Periodic motion ***Both A and B** * None of these
89. The motion of a simple pendulum is S.H.M, if and only if _____.
 *Length of pendulum is small *Mass of pendulum is small
*** Amplitude of motion is small** * All of them
90. Time period of a simple pendulum is independent of _____ pendulum.
 *Mass * Amplitude *Length *** Both A and B**

THE PROFESSOR'S ACADEMY

XI PHYSICS MCQs (CHAPTERWISE)

91. Length of second pendulum on moon is _____m.
*0.165 * .994 *1 * 2.6
92. To obtain time twice the length of pendulum should be _____.
*Doubled * Quartered *Quadrupled * None of them
93. Length of a simple pendulum is increased 44%. Time period of pendulum will be increased _____%
*20 * 30 *40 * 44
94. Time period a simple pendulum will be maximize at _____.
*Pole of earth * Equator of earth *Both A and B * none of them
95. If we cut a spring in to two equal parts, then the spring, constant will _____.
*Remain same * Be doubled * Be Quadrupled * none of them



THE PROFESSOR'S ACADEMY

XI PHYSICS MCQs (CHAPTERWISE)

CHAPTER # 9

31. The appearance of colors in soap bubbles is due to.
(a) Polarization (b) Diffraction
(c) Reflection (d) **Interference**
32. The phenomenon of interference of light was 1st demonstrated by
(a) Newton (b) Einstein
(c) **Thomas Young** (d) Michelson
33. In young's double slit experiment the condition for constructive interference is;
(a) $2d \sin \theta = m\lambda$ (b) **$d \sin \theta = m\lambda$**
(c) $d \sin \theta = \left(m + \frac{1}{2}\right)\lambda$ (d) $d \sin \theta = \left(m + \frac{1}{2}\right)\lambda$
34. When a ray of light enters glass from air then its
(a) Frequency increases (b) Frequency decreases
(c) Wavelength increases (d) **Wavelength decreases**
35. When a ray of light passes from rarer to denser medium, it suffers a phase change of.
(a) 90° (b) **180°**
(c) 60° (d) 45°
36. Wavelength of X-rays is of the order of
(a) $0.1 \text{ \AA}$ (b) **$1 \text{ \AA}$**
(c) $10 \text{ \AA}$ (d) $100 \text{ \AA}$
37. Electromagnetic waves consist of oscillating electric and magnetic fields, both are
(a) Parallel to each other (b) opposite
(c) **Perpendicular to each other** (d) None of these
38. A light beam is entering from one medium to other, which one remains unchanged
(a) Amplitude (b) Velocity
(c) Wavelength (d) **Frequency**
39. The structure of crystal can be studied with the help of _____ x-rays.
(a) Interference (b) **Diffraction**
(c) Polarization (d) Reflection
40. The phenomenon that proves light waves are transfer in nature
(a) Refraction (b) Reflection
(c) **Polarization** (d) Interference
41. Diffraction is a special type of.
(a) Reflection (b) Refraction
(c) **Interference** (d) Polarization
42. In Michelson interferometer, semi silvered plate is used to obtain.
(a) Dispersion (b) **Phase coherence**
(c) Mono chromatic light (d) Unpolarized light
43. Which of the following is not electromagnetic wave?
(a) x-rays (b) Radio waves
(c) Ultraviolet rays (d) **α - rays**
44. Light possesses _____.
(a) Transverse nature (b) Electromagnetic nature
(c) Dual nature (d) **All of these**
45. The number of lines ruled per cm on diffraction grating is 4000. Its grating element is.
(a) **$2.5 \times 10^{-6} \text{ m}$** (b) $2.5 \times 10^4 \text{ m}$
(c) $4 \times 10^6 \text{ m}$ (d) $4 \times 10^{-3} \text{ m}$
46. Light year is a unit of _____.
(a) **Distance** (b) Time
(c) Energy (d) Intensity
47. The dispersion of white light after passing through a prism is due to different _____.
(a) Intensities (b) Amplitudes
(c) Temperatures (d) **Wavelength**

THE PROFESSOR'S ACADEMY

XI PHYSICS MCQs (CHAPTERWISE)

48. To produce bright fringes in Michelson interferometer the moveable mirror is moved through a distance of
- (a) $\frac{\lambda}{4}$ (b) $\frac{\lambda}{2}$
(c) λ (d) 2λ
49. Diffraction of x-rays can be studied by using;
- (a) Thin film (b) Diffraction grating
(c) **Rock salt** (d) None
50. Monochromatic yellow light is unable to show
- (a) Reflection (b) Refraction
(c) **Dispersion** (d) Interference
51. The property of light waves that leads to the phenomenon of color is there _____.
- (a) Velocity (b) Amplitude
(c) Quality (d) **Wavelength**
52. Who proposed electromagnetic wave theory;
- (a) Fresnel (b) Huygen
(c) **Maxwell** (d) Fraunhofer
53. Interferometer measures
- (a) Velocity of light waves (b) Wavelength of mono chromatic light
(c) Thickness of thin objects (d) None of these
54. When electromagnetic waves strike the boundary of denser medium they are;
- (a) Reflected in phase (b) **Reflected out of phase by 180°**
(c) Reflected (d) Completely absorbed
55. In Michelson interferometer semi-silvered mirror is used to obtain
- (a) Thin film interference (b) Phase coherence
(c) Mono chromatic light (d) Colored fringe
56. In diffraction pattern of monochromatic light, the bright bands formed are
- (a) Of uniform intensity (b) **Of non-uniform intensity**
(c) Of uniform width (d) Are of different colors
57. As the order increases, the width of dark band in a diffraction pattern.
- (a) **Increases** (b) Decreases
(c) Does not change (d) Becomes infinity
58. If we narrow the distance b/w two slits in young's experiment the fringes width:
- (a) Increases (b) **Decreases**
(c) Remains the same (d) Becomes zero
59. The condition for interference in thin films is reversed because of
- (a) Small thickness (b) Refraction
(c) **Reflection** (d) Phase reversal
60. Newton's rings are formed due to;
- (a) Interference (b) Diffraction
(c) Polarization (d) **Reflection**
61. Centre of Newton rings always
- (a) Bright (b) **Dark**
(c) Silver (d) None of these
62. When a light enters into a medium then its velocity _____
- (a) **Increase** (b) Decreases
(c) Remains same (d) None of these

THE PROFESSOR'S ACADEMY

XI PHYSICS MCQs (CHAPTERWISE)

CHAPTER # 10

1. If a single convex lens is placed closed to eye, it is used as;
(a) Compound microscope (b) Telescope
(c) **Simple microscope** (d) Spectrometer
2. The power of concave lens of focal length 50cm is
(a) 0.5 diopter (b) **-2 diopter**
(c) 2 diopters (d) 5 diopters
3. The unit of power of lens is
(a) Walt (b) Joule
(c) **Diopter** (d) Newton
4. If focal length of magnifying glass is 5cm, its magnification is
(a) 2 (b) **6**
(c) 4 (d) 20
5. A defect of eye called myopia can be corrected by using the _____.
(a) **Convex lens** (b) Concave lens
(c) Bifocal lens (d) None
6. If an object is placed at 2F of a convex lens, then image formed at;
(a) **2F** (b) 4F
(c) 3F (d) F
7. Chromatic aberration is caused by;
(a) Reflection (b) **Dispersion**
(c) Refraction (d) None of these
8. In an astronomical telescope of focal length of objective is 90 cm eye piece focal length is 10 cm, the length of telescope is
(a) 9 cm (b) **100 cm**
(c) 80 cm (d) 0.9 cm
9. The length of Galilean telescope is
(a) $\frac{f_o}{f_e}$ (b) **$f_o - f_e$**
(c) $f_e - f_o$ (d) $f_o + f_e$
10. If an object is placed at principal focus F of a converging lens, the image will be formed at
(a) F (b) 2F
(c) **At infinity** (d) All of these
11. The final image formed by a compound microscope is
(a) **Virtual and diminished** (b) Real and diminished
(c) Red and magnified (d) Virtual and magnified
12. The distance b/w the principal focus and optical center is called
(a) Aperture (b) Radius of curvature
(c) Principal axis (d) **Focal length**
13. In convex lens when the object is placed beyond 2F then its image will form;
(a) **At 2F on the other side** (b) b/w F and 2F on the other side
(c) Beyond 2F on the other sides (d) All these
14. To get higher magnification by magnifying glass, the lens is used of;
(a) **Short focal length** (b) Large focal length
(c) None of these (d) Both a and b
15. The length of diverging lens is 6cm and object is placed at 12cm from it then image formed at a distance of
(a) 2 cm (b) **-4 cm**
(c) -6cm (d) 8cm
16. In Galilean telescope the final images is
(a) Red and magnified (b) **Virtual and magnified**
(c) Real and diminished (d) Virtual and diminished

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17. A telescope has objective of focal length 60cm and eye piece of focal length 3cm then its magnifying power is
(a) 2 (b) **20**
(c) 180 (d) 57
18. Under normal condition the human eye is most sensitive for;
(a) Red light (b) **Yellow green light**
(c) Blue light (d) Purple light
19. The diameter of inner rim of a lens is called
(a) Circumference (b) **Radius of curvature**
(c) Aperture (d) Focal length
20. Focal length of plane glass is _____.
(a) Zero (b) **Infinite**
(c) 25 (d) Not fixed
21. Power of plane glass is
(a) **Zero** (b) Infinite
(c) 25 (d) Not fixed
22. Twice of focal length is equal to _____.
(a) **Radius of curvature** (b) Power of lens
(c) Magnifying power (d) Apertures
23. Convex lens form virtual image when the object is placed
(a) At F (b) At 2F
(c) Beyond 2F (d) **With in F**
24. Image is formed at focus by a convex lens when the object is placed;
(a) At F (b) At 2F
(c) Beyond 2F (d) **At infinity**
25. Object is placed beyond 2F of a convex lens the image is formed;
(a) At F (b) At 2F
(c) Beyond 2F (d) **Between F and 2F**
26. The image formed by a convex lens at its focus, the image is
(a) **Real and diminished** (b) Virtual and diminished
(c) Real and magnified (d) Virtual and diminished
27. Image of an object formed by convex lens, big in size, the object is
(a) **b/w F and 2F** (b) At 2F
(c) Beyond 2F (d) At infinity
28. Compound microscope consists of _____ lenses;
(a) Two convex (b) **Two concaves**
(c) Convex and concave (d) All these
29. In compound microscope focal length of objective is always _____. Then the focal length of eye piece
(a) Greater (b) **Less**
(c) Equal (d) Twice
30. Magnification of objective is 6 and eye piece is 10, the total magnification of compound microscope is;
(a) 4 (b) 16
(c) **60** (d) 0.6
31. _____ lens act as a magnifying glass in compound microscope
(a) Objective (b) **Eye piece**
(c) Both (d) None of these
32. Final image formed by a compound microscope is
(a) Real, erect, Magnified (b) Magnified, Real, inverted
(c) **Virtual, erect, magnified** (d) Magnified, virtual, inverted
33. Final image formed by the astronomical telescope is at
(a) **Infinity** (b) At focus of objective
(c) At focus of eyepiece (d) Least distance of distinct vision
34. The mirrors which are used in searchlights and spotlights are
(a) Plane mirrors (b) **Concave mirrors**
(c) Convex mirrors (d) All these

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35. Whenever high magnification is desired

(a) Simple microscope is used

(c) **Compound microscope is used**

(b) Convex lens is used

(d) Lenses are used in combination

